Reinforcement-Learning Framework for Robust and Universal Analog Quantum Control on Superconducting Qubits

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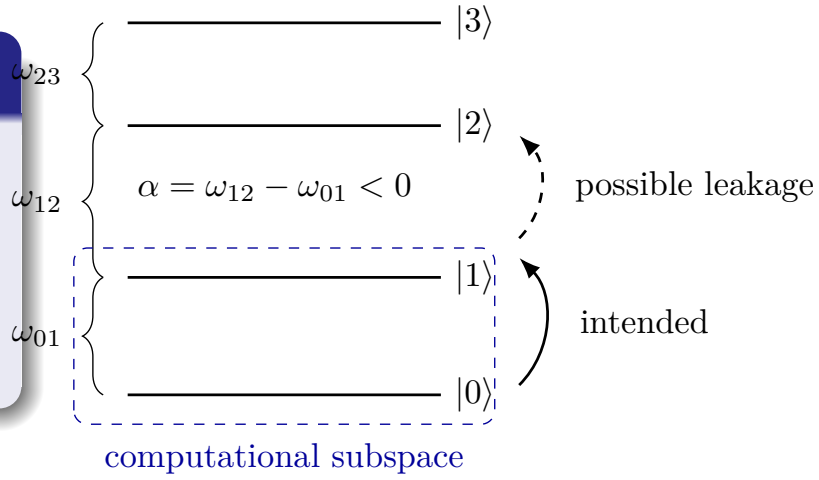
Why this problem matters

Main difficulty

In superconducting qubits, gate design is not only a matter of logical accuracy. Fast pulse-level control must balance several competing goals:



Computational subspace vs leakage subspaces.

- **high gate fidelity**
- **short runtime**
- **low leakage** outside the computational subspace
- **robustness** against control noise

Physical issue

Superconducting qubits are only **weakly anharmonic**, so fast control can easily excite

Thesis goals and main contribution

Scientific goal

To assess whether the framework proposed by Niu *et al.* is a credible strategy for **robust analog quantum control**.

Important point

This was not a standard reproduction from an existing codebase.

What I specifically did

- reconstructed the full framework from the paper
- implemented simulation, leakage estimation, and RL pipeline
- performed an independent empirical validation
- compared nominal, noisy, and family-wide behavior

- No public implementation was available
- The whole framework had to be rebuilt from scratch
- This included:
  - multilevel quantum simulation
  - TSWT-based leakage estimator
  - TRPO training pipeline
  - robustness and runtime experiments

Physical model: gmon Hamiltonian and control channels

The effective bosonic Hamiltonian is

$$\hat{H}_{\mathrm{RWA}}(t) = \frac{\eta}{2}\sum_{j=1}^{2}\hat{n}_j(\hat{n}_j-1)+g(t)(\hat{a}_2^\dagger\hat{a}_1+\hat{a}_1^\dagger\hat{a}_2)+\sum_{j=1}^{2}\delta_j(t)\hat{n}_j$$

The control vector is

$$u(t) = \big(g(t),\, \delta_1(t), \delta_2(t),\, f_1(t), f_2(t),\, \phi_1(t), \phi_2(t)\big).$$

After three-level truncation:

$$d_{\mathrm{phys}} = 3^2 = 9, \qquad d_{\mathrm{comp}} = 4.$$

Meaning of the controls

- $g(t)$: tunable inter-qubit coupling
- $\delta_j(t)$: local detunings
- $f_j(t)$: microwave amplitudes
- $\phi_j(t)$: microwave phases

Why this matters

This is a **continuous 7-dimensional control problem** on a multilevel superconducting system.

Leakage treatment and control objective

To control leakage, the paper uses a **time-dependent Schrieffer–Wolff transformation**.

The resulting leakage proxy is

$$L_{\mathrm{tot}}(T) \lesssim \frac{\|\hat{H}_{\mathrm{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\mathrm{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\mathrm{od}}(t)}{dt^2} \right\| dt.$$

The full UFO objective is

$$C = \chi[1 - F(U(T))] + \beta L_{\mathrm{tot}} + \mu P_{\mathrm{bdry}} + \kappa T.$$

Suggested figure:

`ufo_cost_structure.pdf`

Interpretation

The optimizer must balance: fidelity, leakage, pulse regularity, and runtime.

RL formulation and target gate family

Target gate family

$$N(\alpha, \alpha, \gamma) = \exp[i\,(\alpha\,\sigma_1^x\sigma_2^x + \alpha\,\sigma_1^y\sigma_2^y + \gamma$$

This is the family used throughout the numerical study.

- continuously parametrized by α, γ
- physically meaningful for the gmon interaction structure
- suitable for both single-target and family-wide analysis

RL formulation

- **state**: current propagated unitary U_t
- **action**: next continuous control vector
- **environment**: noisy multilevel quantum dynamics
- **reward**: negative UFO cost
- **algorithm**: TRPO actor–critic

Why RL here?

Because the problem is sequential, continuous, and explicitly multi-objective.

Evaluation metrics

Metric	Definition	Physical meaning
Gate fidelity	$F_{\text{gate}} = \frac{1}{16}\left\lvert \mathrm{Tr}\left(U_{\text{proj}}^{\dagger}(T)U_{\text{target}}\right)\right\rvert^2$	Logical accuracy of the implemented two-qubit gate.
Leakage proxy	L_{tot}	Estimated coherent population loss outside the computational subspace.
Average fidelity under noise	$\overline{F}(\sigma) = \mathbb{E}[F]$	Mean gate performance over many noisy Monte Carlo realizations.
Fidelity variance	$\mathrm{Var}(F) = \mathbb{E}\left[(F - \overline{F})^2\right]$	Shot-to-shot stability under stochastic perturbations.
Runtime	T	Hardware efficiency: how long the gate takes to execute.

Framework reconstruction from scratch

What had to be implemented

- multilevel gmon simulator
- bandwidth-limited control filtering
- TSWT-based leakage estimator
- RL environment and TRPO agent
- training, evaluation, and robustness pipelines
- runtime sweeps across the gate family

Put here your framework image

(the one you sent me),

or fallback to

`paper_rl_loop.pdf`

Why this is relevant

The thesis contribution is not only empirical testing: it includes the **full reconstruction** of a nontrivial hybrid

Extended parameter study

This part was essential

In RL, parameter choice is not a minor detail: it strongly affects stability, convergence, robustness, and final control quality.

I performed an extended study on:

- **simulation/training parameters**
  - $\Delta t = 2\,\mathrm{ns}$, $T_{\mathrm{max}} = 180\,\mathrm{ns}$
  - $\log \sigma_0 = -1$, $\delta_{\mathrm{KL}} = 0.01$
  - $\sigma_{\mathrm{train}} = 1\,\mathrm{MHz}$
- **policy/value architectures**
  - sweep over 5 network topologies
  - final choice: $(64, 32, 32)$, paper-faithful and competitive
- **UFO weights**

Suggested figure:

`ufo_weight_sweep_pareto.pdf`

or

`architecture_sweep_pareto_co`

Single-target benchmark: Adam vs nominal TRPO

Representative target:

$$N(2.2, 2.2, \pi/2)$$

Metric	Adam	Nominal TRPO
Runtime	70 ns	116 ns
Cost	0.1576	0.2982
Fidelity	0.99662	0.99415
Leakage	2.82×10^{-4}	1.60×10^{-3}

Suggested figure:

`single_target_adam_vs_nominal_trp`

Interpretation

In the purely nominal single-target regime, the direct Adam baseline remains stronger.

Noise-optimized TRPO: representative training dynamics

What this plot shows

A representative training run of the learned controller:

- evaluation cost
- execution time
- fidelity
- leakage

Interpretation

Training is not monotonic: the policy must learn how to balance fidelity, leakage, and runtime at the same time.

Put here the training-curves figure of the
noise-optimized run.

Current fallback:
`nominal_trpo_training_curves.pdf`

Robustness under stochastic control noise

Main empirical result

Under Gaussian control noise, the **noise-optimized TRPO controller** becomes the strongest overall method.

- highest $\overline{F}(\sigma_{\text{noise}})$
- lowest $\mathrm{Var}(F)$
- better than:
  - nominal TRPO
  - Adam baseline

avg_fidelity_vs_noise.pdf

fidelity_variance_vs_noise.pdf

Take-home message

The advantage is not just RL itself, but **RL trained in the correct stochastic environment**.

Runtime across the gate family $N(\alpha, \alpha, \pi/2)$

$$U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2)$$

- The runtime landscape is **highly structured**, not uniform.
- A broad central region admits significantly faster analog solutions.
- The analog controller stays below the gate-synthesis reference over a large part of the family.

Suggested figure:

`runtime_vs_alpha.pdf`

Reference benchmark

$$T_{\text{syn}} \approx 215\,\text{ns}$$

from optimal discrete gate synthesis.

Physical interpretation

Conclusions

Scientific conclusions

- The framework is physically meaningful and empirically credible.
- In the nominal single-target regime, direct pulse optimization remains very strong.
- Under stochastic perturbations, noise-optimized RL becomes the bes strategy.
- Analog control can outperform standard gate synthesis in runtime.

What this thesis contributed

- full reconstruction of the framework from scratch
- implementation of simulation, leakage, and RL pipeline
- extended parameter study
- independent validation of strengths and limitations

Final message

The main value of this framework is not universal nominal dominance, but a **unified strategy for robust, leakage-aware, hardware-relevant analog quantum control**